

Electrical Line Worker Safety Training

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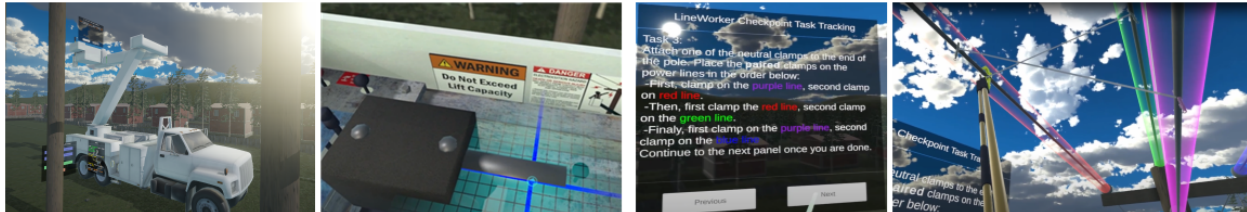


Figure 1. The experimental environment, the handle to move the bucket, and using the instructions to connect the lines

Abstract – Safety training in any environment is a necessity that most, if not all companies use in order to provide a boundary where tests can be made without incident. We present the concept and overall idea of implementing real-world classroom training in a VR simulation specified for Electrical Line Workers. Our simulation will include four scenarios used to test the ability of each participant and the information they retain by providing a questionnaire and a NASA TLX survey of the workload. Our goal is to provide a safety simulation that training facilities can use to provide a hands-on experience to measure the success of their electrical worker trainees.

Keywords–Virtual Reality, Presence, Line Worker Training, Training, Simulation, Safety

Introduction

Virtual Reality (VR) is used in multiple settings to expand on existing research and training to understand many different fields within a boundary of safety. Although most VR applications are known from the games industry, more emphasis has been developed to better understand where technology can go when increasing the number of resources in our society [6].

Many programs around the country focus on a classroom-based learning environment before going into the field which only has a word-of-mouth approach instead of a hands-on approach. TEEX (Texas A&M Engineering Extension Service) is

currently looking for ways to improve their training environment that can have a more effective approach to successful outcomes and compliments their already taught curriculum [4]. The simulation includes four scenarios that TEEX can use from classroom training in a VR simulation specified for Electrical Line Workers [20].

Creating a VR application for the understanding of Linework training provides a safe environment that can be used as both a recruiting mechanism and a solid foundation for learning. Our focus is to understand the user's correctness when given instructions and colored indicators in the environment; this also includes the interaction of height, sound, and touch in the simulation [5].

Background

What Is An Electrical Line Worker?

An Electrical Line Worker (also known as a Line Worker) is someone who directly maintains the electric transmission and distribution facilities that deliver electricity to residential, commercial, and industrial establishments [12]. Their main focus is to restore, install, and provide service to any electrical lines needed. Many cases like severe weather or damage require a Lineworker to service a line [19]. Electrical Line Worker training for new hires at TEEX is currently a classroom-based training approach used to teach the main topics needed to successfully work with electrical wires and connections.

VR Used for Professional Training

VR has been used as a tool for training across diverse settings with varying levels of success in the past [1]. VR training is a great source of understanding a different approach of training and the success of that training [13]. Industrial training is one of the major application domains for VR, as it allows to learn and perform simulated hands-on activities in a controlled, safe environment [1]. Many studies rely on a safe foundation that can engage users to learn in a positive environment with motivation and desire under instructional goals. This rationale was approached and underpinned by applying the input-process-output model of instructional game characteristics matched to specific learning outcomes [2]. Education and training awareness is a serious development many companies consider when hiring new.

VR used for Lineworker Training

The electrical power industry is just one industry in need of different forms of training for a more intensive and successful experience. These simulations provide many advantages to different individuals and their learning habits [7]. With requirements already in-hand for all construction and electrical workers there is a responsibility required of each individual. [8] These requirements can be life threatening because of the accidents that happen with live lines making it lethal [9]. It's important for companies to try different sources of training for new electrical workers because of the health risks and challenges many face [14]. Safety programs are an important part of training because inexperienced workers fail to recognize the potential injury associated with electrical work when performing a task [10]. Electrical wires pose more threats than many inexperienced workers may know [11].

Methodology

Our project was created using the software Unity. We've pulled assets from the Unity Marketplace in order to help us achieve a real-world environment. Our final project will be used in the Meta Quest 2 headset powered by Unity [18]. ([GitHub Link](#)) Our solution to Electrical Line Worker Training is to provide the steps needed in order to successfully complete any task. The tasks in our simulations are listed as: a) Step On the Bucket Truck, b) Look Over

the Edge of the Bucket Truck, c) Bucket Truck Operation, d) Measuring Voltage, e) Placing Cables on the Lines.

Experimental Apparatus

Experimental Scenarios

Our simulation consists of four scenarios all with the same actions required to complete the training. Below is a list of the four scenarios: (a) Instructions with Indicators, (b) Instructions Only, (c) Indicators Only, (d) No Instructions or Indicators. We will have a total of eight experimental scenarios with the inclusion or removal of environmental effects.

Instructions will be used to explain what each of the required tasks are. These will be present at the beginning of the simulation and throughout each step. Indicators are the highlighted assets that the trainee will need to use in order to complete the task. Our different scenario options will be at the beginning of the simulation before the trainee enters the bucket truck.

Below we have a list of tasks required for the user in order to complete the simulation (Instructions Only Scenario):

1. Step On The Bucket Truck: Trainees will be required to use the handles on the truck to enter the truck with the grip button.
2. Look Over The Edge Of The Bucket: Trainees will need to look over the edge of the bucket to evaluate the bucket's location.
3. Bucket Truck Operation: Trainees will then use the controllers in the bucket to correctly position the bucket under the lines in order to interact with them correctly. (Instructions will guide the trainee on where to place the bucket).
4. Measuring Voltage: Trainees will then use the multi-purpose pole (cable tester attached) to check the line's voltage.
5. Place Cables On Lines: The final step is to place a cable on the end of the pole, put it in position, and add it to the line. (An indicator will appear to show the range in which the cable can be placed.) Once placed, the other end of the cable will then need to be placed on the line.
6. Completion: Trainees will then hit the button on the interface to mark the simulation as finished.

Performance Testing

In order to properly gain the right information needed to successfully continue on to real-world training, all users will be introduced to an environment where the scenarios listed above provide further environmental factors. One of the main factors used to test the ability of someone being a Lineworker is the heightened sensation of being high above the ground in a bucket truck. This includes the experience of wind rustling audio, distorted ground, and a wobble in the bucket.

To evaluate the performance under the different conditions, we tracked each task completion in the simulation, and provided information on whether a task was completed fully, incorrectly, or not finished. We specifically tested for the correct placement of the bucket, usage of the power testing tool, and the hook placement order on the lines. To get additional information, we timed the experiment and tracked uses of the bucket handle plus the number of times the user dropped tools. These metrics are meant to aid in finding trends that reveal patterns in users' ability to avoid errors.

Post Simulation Questionnaires

All users will be required to fill out our User Experience Questionnaire. Some of the questions may include: How did you feel in the bucket? How difficult were the tasks? How often did you feel unsure of yourself? A second set of questions will be provided in our NASA Task Load Index Survey (NASA TLX). The purpose of this survey allows us to capture a quantitative measure of the simulation's workload [15]. Lastly, a third set of questions focused on an IGroup Questionnaire to identify the effects in environmental changes based on immersion and presence.

Participant Information

We have a total of 16 participants (8 females, 7 male, and 1 other). All participants were between the ages of 20-75 with the majority on the young side. Only 5 out of 16 have a higher ranking in VR experience.

Results

Heights

Line worker tasks require all trainees to work with heights; 93.8% were unafraid of heights during the simulation. 81.4% of all users claimed to NOT be

afraid of heights, while 6.3% were neutral, and 12.5% were afraid of heights.

Tasks

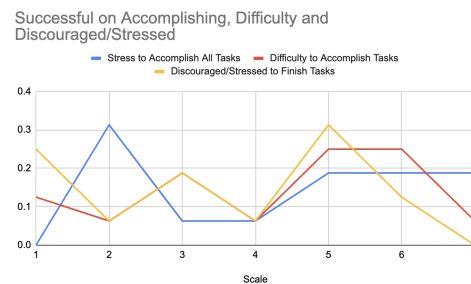


Fig. 2. Perceptions of performance, difficulty, and stress

Figure 2. shows the results on a scale of 1 to 7 on how difficult it was to accomplish tasks based on performance, difficulty, and stress. All users were asked multiple questions on a scale of 1 to 7 based on their experience with understanding the tasks and the demands each task required. Based on the mentality (required thinking) of the tasks 31.4% didn't think they were demanding while 12.5% were neutral, and 56.3% found the tasks highly stressful. 62.6% found all tasks to be less physically demanding which shows us that more thought was required than movement. 75.1% of all participants agreed that they did not feel rushed or hurried to complete all tasks. Overall, 25.1% felt sure and confident of completing tasks while 12.5% felt neutral, and 62.5% of all participants felt much difficulty with completing all tasks successfully.

Environment

While completing all tasks 100%, of participants felt as if they were immersed in the environment. 75% of participants also felt as if they were operating something in the simulation versus 6.3% feeling they were operating something outside [17].

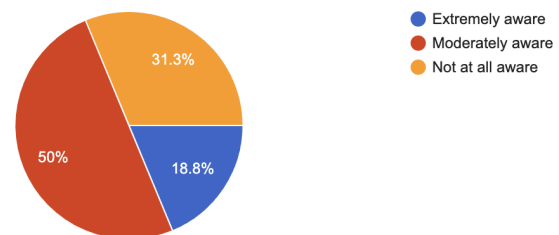


Fig. 3. Participants' awareness of outside environment

Figure 3. further explains how aware each participant was of their surrounding environment within the

simulation, showcasing that only 18.8% were aware of real-world surroundings. Lastly, 100% of participants felt immersed by the virtual world with 53.4% sharing that it didn't feel like the real world [16].

Objective Metrics

These results consist of details needed to see how successful each participant really was within the environment. Within the four different scenarios 50% of participants used all environmental factors. 6 participants used Instructions and Indicators, 4 used Instructions only, 3 used Indicators only, and 3 used neither. The time each participant took to complete a scenario ranged from 212 seconds to 1740 seconds. Within the tasks, a range of tool drops went from 0 to 22 and handle grabs from 0 to 21. Five tool drop counts were invalid due to an error corrected in the subsequent trials.

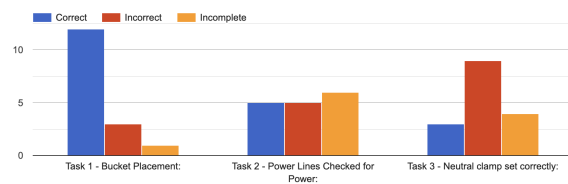


Fig. 4. Task Completion Charts

Figure 4. shows the number of participants who were able to complete each task and how successful they were to get to the end of the final task (far right of the chart).

Discussion

We can confirm that test results range from participant to participant based on experience and knowledge of the tasks being asked [3].

Type	Participant 1	Participant 2	Participant 3	Participant 4
Gender	Male	Female	Male	Male
Experience with VR	1	1	6	7
Lost or Unsure	1	4	5	5
Felt Heights	7	7	6	6
Afraid of Heights?	No	No	Yes	No
Mental Demand	4	6	3	5
Physical Demand	2	1	4	5
Hurried / Rushed	4	1	3	3
Feeling of Success	2	3	4	2
Felt Stressed/Discour	1	1	3	6
Scenario	Indicat...	Neither	Neither	Indicat...
Time (seconds)	417	1080	434	504
Tool Drops	3	2	3	177
Task 1	Compl...	Compl...	Compl...	Compl...
Task 2	Compl...	Compl...	Compl...	Incom...
Task 3	Compl...	Incorrect	Incorrect	Compl...

Fig. 5. Comparison of low and high VR experience participants

Within all four participants shown in Figure 5, even those with more experience still struggled with task completion. Therefore, we believe that our VR simulation will complement the existing curriculum used in training. However, the participants in our research came from outside of the TEEX program.

Participants with some of the highest task completion scores still expressed feeling lost or confused with the tasks and had no prior experience with VR. We hypothesize that part of their confusion came from a lack of understanding of the technology itself. This is further backed up by no real trend with the type of scenario used and the level of understanding. We may be able to resolve this in the future by providing a more in-depth tutorial for those that are new to the technology.

Overall, the first task of moving the bucket was completed the most, and the second task of measuring voltage was completed the least. We believe that anyone who completed the Instructions and Indicators scenario first would have more success with every other scenario afterward. This will provide training facilities with more information on where each trainee is at, and how well they performed with hands-on experiences.

Conclusion

We have presented the concept and overall idea of implementing real-world classroom training in a VR simulation specified for Electrical Line Workers. Our simulation focused on four different scenarios that were used to test the ability of each trainee and the information retained. The results of our study provided significant findings that will aid in future works. Our measurements have helped us to determine where trainees could be when training and where they may need help to successfully graduate and move on into the field. Our results had a valuable outcome of helping facilities understand the training needed to help those coming into the field, and where they can go to further the understanding needed for training.

Sources

- [1] S. kumar Renganayagalu, S. C. Mallam, and S. Nazir, "Effectiveness of VR head mounted displays in professional training: A systematic review - technology, knowledge and learning," *SpringerLink*, 01-Jan-2021. [Online]. Available: <https://link.springer.com/article/10.1007/s10758-020-09489-9>. [Accessed: 05-May-2023].
- [2] Author links open overlay panelJack Goulding a, a, b, c, d, AbstractThe 'traditional' construction industry has constantly been challenged to improve its inherent problematic practices. Offsite production (OSP), S. Defreitas, B. de Vries, S. Leitch, S. Gurjao, W. Nadim, D. Leabue, A. Gibb, L. Clarke, R. Buckley, G. R. Amthor, B. Koo, and M. Rossou, "Construction industry offsite production: A virtual reality interactive training environment prototype," *Advanced Engineering Informatics*, 17-Nov-2011. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1474034611000851>. [Accessed: 06-May-2023].
- [3] S. W. Greenwald, A. Kulik, A. Kunert, S. Beck, B. Fröhlich, S. Cobb, S. Parsons, N. Newbutt, C. Gouveia, C. Cook, A. Snyder, S. Payne, J. Holland, S. Buessing, G. Fields, W. Corning, V. Lee, L. Xia, and P. Maes, "Technology and applications for collaborative learning in virtual reality," *Repository of the International Society of Learning Sciences: Technology and Applications for Collaborative Learning in Virtual Reality*, 01-Jul-2017. [Online]. Available: <https://repository.isls.org/handle/1/210>. [Accessed: 07-May-2023].
- [4] A. authors, P. Smutny, and Additional informationFundingThis work was supported by the project SP2022/60 Applied Research in the Area of Machines and Process Control supported by the Ministry of Education, "Learning with virtual reality: A market analysis of educational and training applications," *Taylor & Francis*. [Online]. Available: <https://www.tandfonline.com/doi/full/10.1080/10494820.2022.2028856>. [Accessed: 07-May-2023].
- [5] M. North, J. Hill, A. Aikhuele, and S. North, "Virtual reality training in aid of communication apprehension in classroom environments," *International Journal of Emerging Technologies in Learning (iJET)*, 03-Jun-2008. [Online]. Available: <https://www.learntechlib.org/p/45328/>. [Accessed: 07-May-2023].
- [6] B. Xie, H. Liu, R. Alghofaili, Y. Zhang, Y. Jiang, F. D. Lobo, C. Li, W. Li, H. Huang, M. Akdere, C. Mousas, and L.-F. Yu, "A review on Virtual reality skill training applications," *Frontiers*, 19-Apr-2021. [Online]. Available: <https://www.frontiersin.org/articles/10.3389/frvir.2021.645153/full>. [Accessed: 07-May-2023].
- [7] "Using simulations to provide safety and operations training in the ..." [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/747333>. [Accessed: 07-May-2023].
- [8] "Using virtual environments to support electrical safety ... - IEEE xplore." [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/5429258/>. [Accessed: 07-May-2023].
- [9] A. Ayala García, I. Galván Bobadilla, G. Arroyo Figueroa, M. Pérez Ramírez, and J. Muñoz Román, "Virtual reality training system for maintenance and operation of high-voltage overhead power lines - virtual reality," *SpringerLink*, 13-Jan-2016. [Online]. Available: <https://link.springer.com/article/10.1007/s10055-015-0280-6>. [Accessed: 07-May-2023].
- [10] "From the 'ground' up improving electrical safety for the electrical ..." [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/7914855/>. [Accessed: 07-May-2023].
- [11] D. Wallis and B. Erga, "Protecting Workers from Electromagnetic Induction During Line-Stringing Operations: Copyright Material IEEE Paper No. ESW2021-02," *2021 IEEE IAS Electrical Safety Workshop (ESW)*, Tucson, AZ, USA, 2021, pp. 1-5, doi: 10.1109/ESW45993.2021.9461531. [Accessed: 07-May-2023].
- [12] Author links open overlay panelR Aracil a, a, b, AbstractThis paper is focused on a telerobotic application for the maintenance of electrical live-power lines, S. Bonner, and J. Coté, "Telerobotic

system for live-power line maintenance: Robtet,” *Control Engineering Practice*, 15-Nov-2002. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S096706610200182X>. [Accessed: 07-May-2023].

[13] “Virtual reality for education and workforce training | IEEE conference ...” [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/8102472>. [Accessed: 07-May-2023].

[14] Author links open overlay panelVitaly Volberg a, a, c, b, d, e, AbstractIntroductionWorkers in the electric power industry face many risks of injury due to the high diversity of work tasks performed in potentially hazardous and unpredictable work environments.MethodWe calculated injury rates by age, G. S. Sorock, M. K. Jones, T. A. Fordyce, T. Braun, P. Brooks, M. Capelli-Schellpfeffer, N. Chau, B. N. Craig, Epri, A. Farrow, J. L. Fleiss, and H. L. Floyd, “Injuries among electric power industry workers, 1995–2013,” *Journal of Safety Research*, 25-Nov-2016. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S0022437516304200>. [Accessed: 07-May-2023].

[15] C. Nikulin, et al., "NASA-TLX for predictability and measurability of instructional design models: case study in design methods," *Educational Technology Research and Development*, vol. 67, pp. 467-493, 2019.

[16] “Virtual reality: How much immersion is enough? | IEEE journals ...” [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/4287241/>. [Accessed: 07-May-2023].

[17] “Virtual/reality: How to tell the difference - sage journals.” [Online]. Available: <https://journals.sagepub.com/doi/pdf/10.1177/1470412920906253>. [Accessed: 07-May-2023].

[18] “Developing virtual reality applications with unity | IEEE conference ...” [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/6802117>. [Accessed: 07-May-2023].

[19] A. Pagnano, M. Höpf, and R. Teti, "A roadmap for automated power line inspection. Maintenance

and repair," *Procedia Cirp*, vol. 12, pp. 234-239, 2013.

[20] Additional informationFundingThis work was supported by National Natural Science Foundation of China [grand numbers 71201155; 71571133]., “How virtual reality affects perceived learning effectiveness: A task–technology fit perspective,” *Taylor & Francis*. [Online]. Available: <https://www.tandfonline.com/doi/abs/10.1080/0144929X.2016.1268647>.

[GitHub Link](#)